

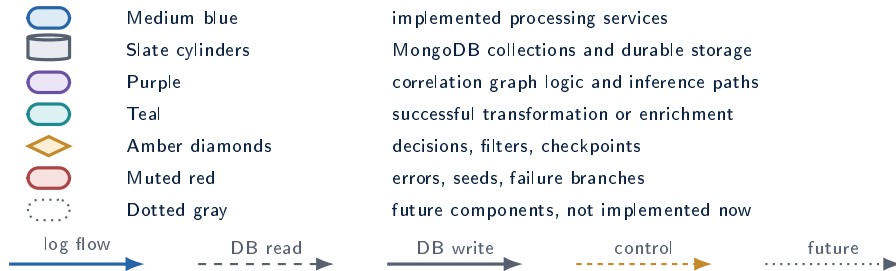
OpenStack RCA Copilot

Evidence-first, graph-first root cause investigation for OpenStack

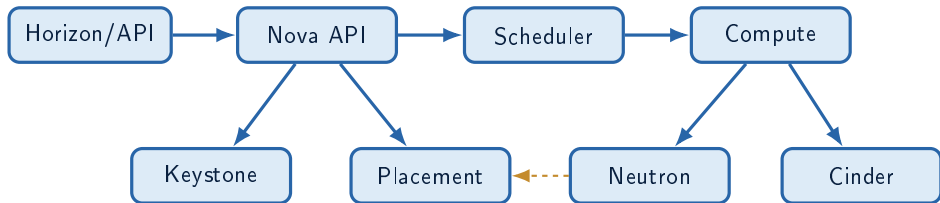
Project Presentation

July 6, 2026

Semantic Visual Language

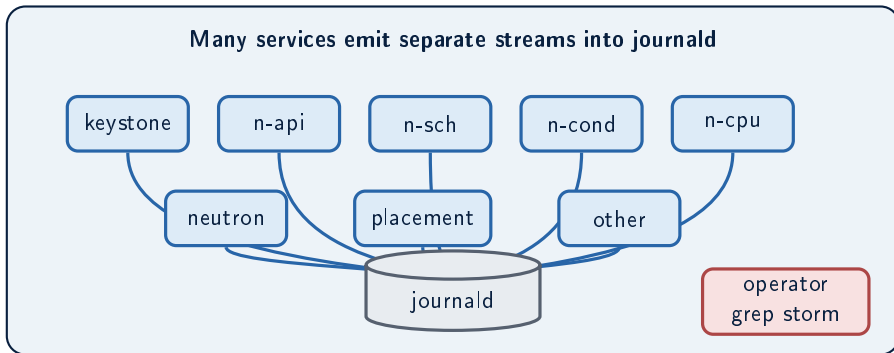


OpenStack Service Ecosystem

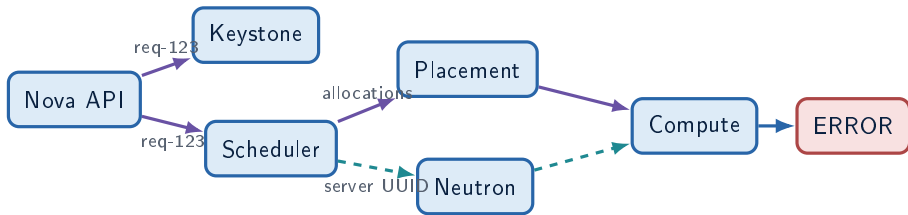


A single operation crosses independent services, logs, IDs, and clocks.

Distributed Log Volume

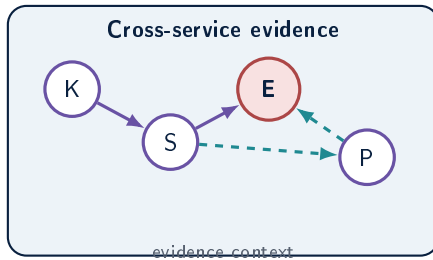
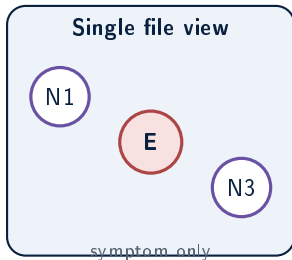


One Request Crosses Many Services

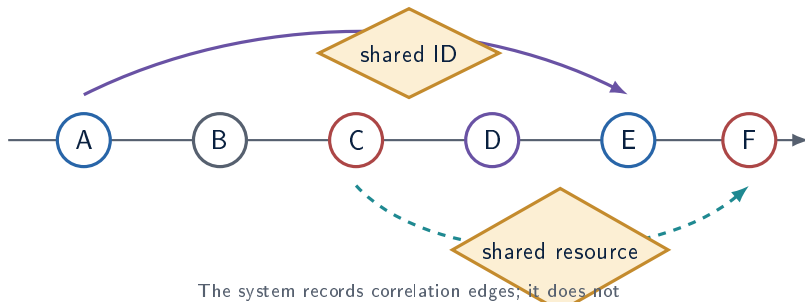


The visible failure may be far from the first relevant event.

Why Isolated Log Inspection Fails

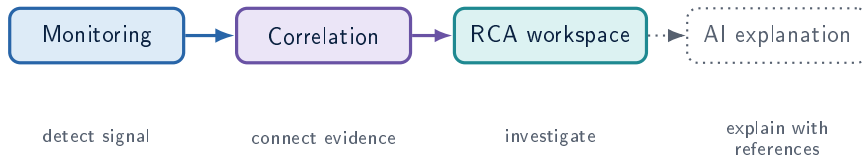


Chronology Is Not Causality

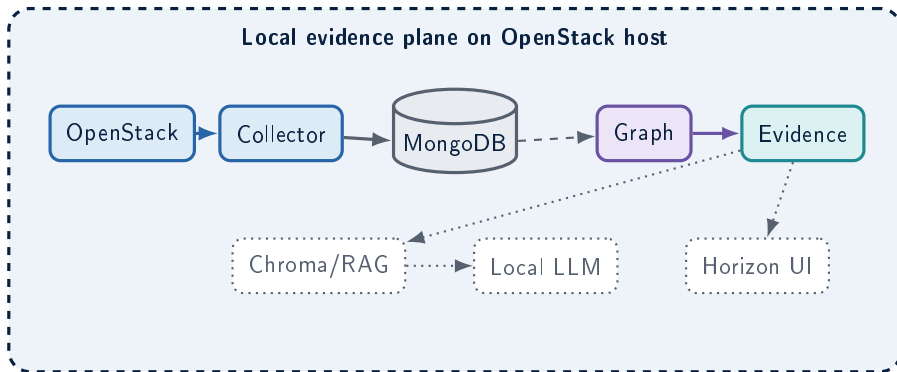


The system records correlation edges; it does not claim root cause until future grounded analysis.

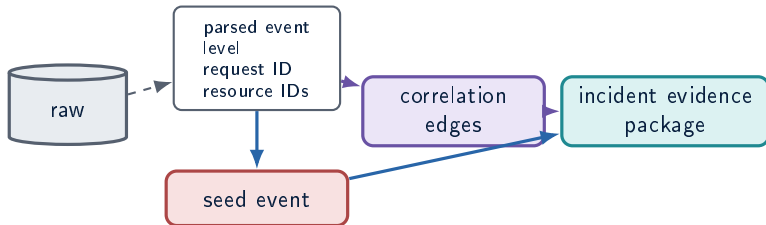
Monitoring vs Correlation vs RCA vs AI



Vision Architecture

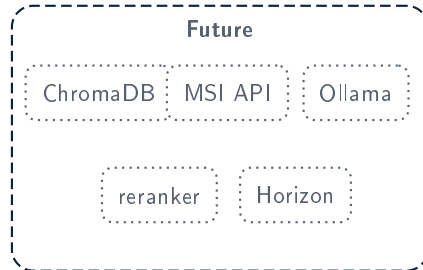
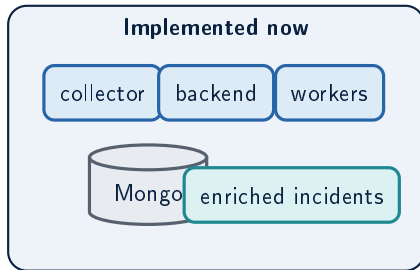


Evidence-First Design

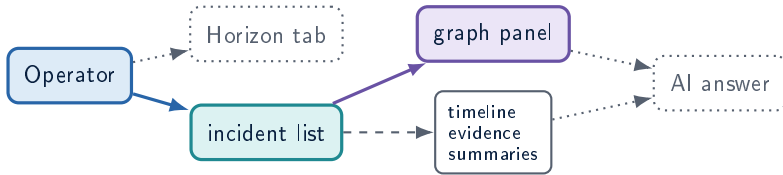


AI is downstream of evidence, never a substitute for collected facts.

Implemented vs Future System

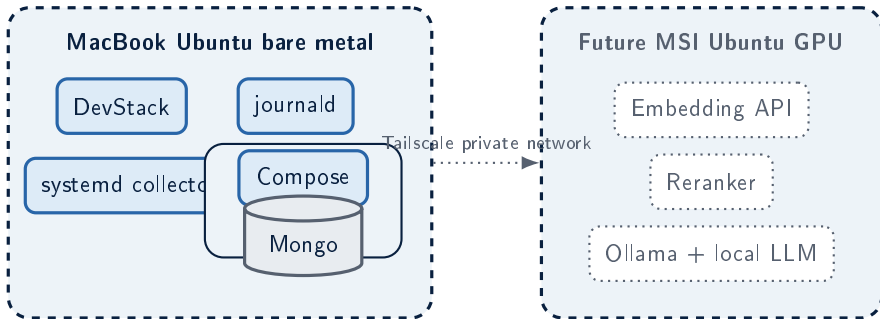


Operator Interaction Model

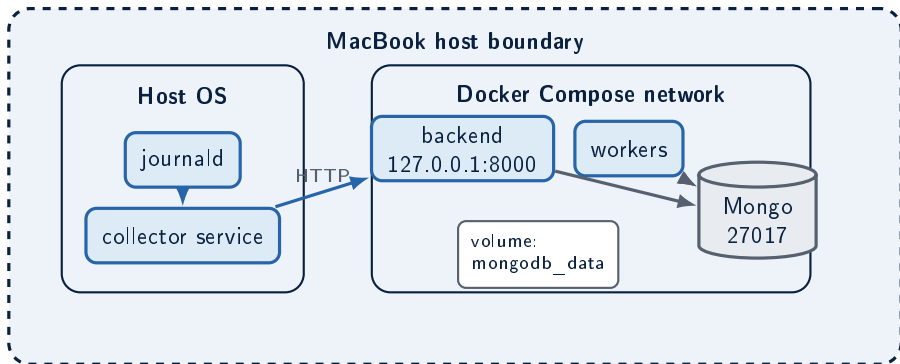


The browser never calls Ollama directly; backend mediates future inference.

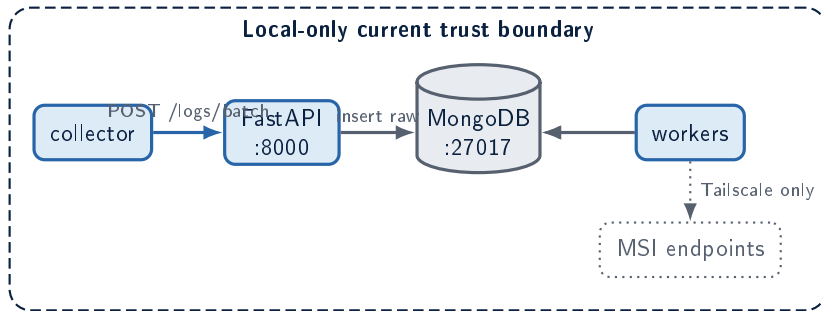
Two-Machine Physical Topology



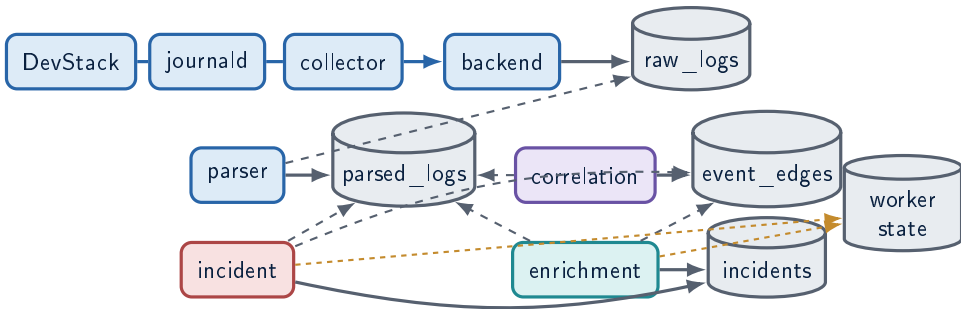
MacBook Host and Container Boundaries



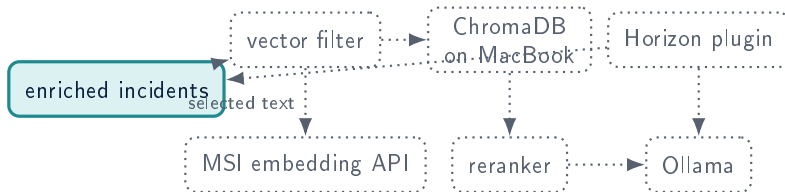
Ports, Protocols, and Trust Boundaries



Runtime Dependency Graph

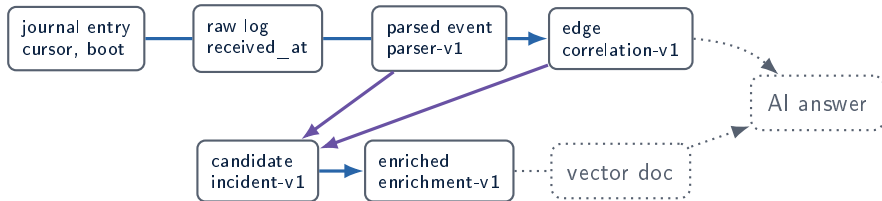


Future Runtime Extensions

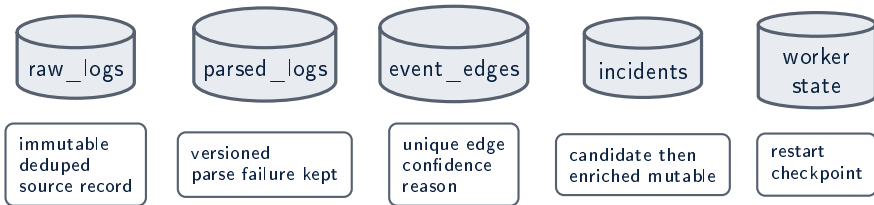


MongoDB remains source of truth; Chroma stores vectors and metadata only.

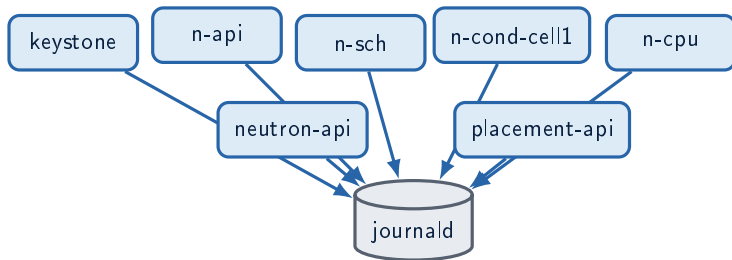
Data Lineage Map



Artifact Responsibilities

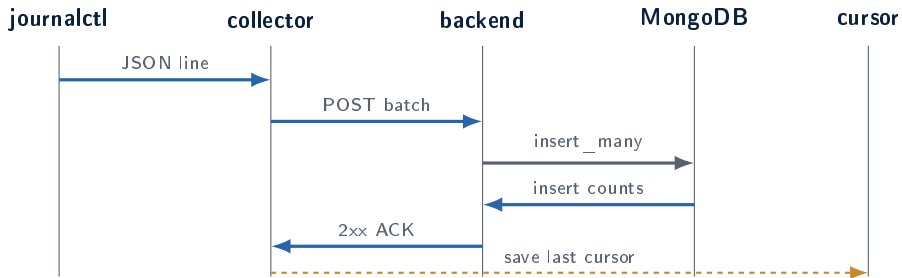


Monitored DevStack Units

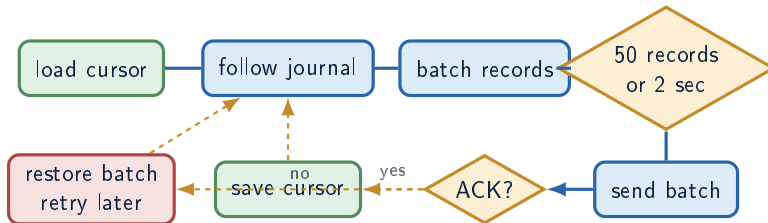


Default unit list from `collector/config.py`

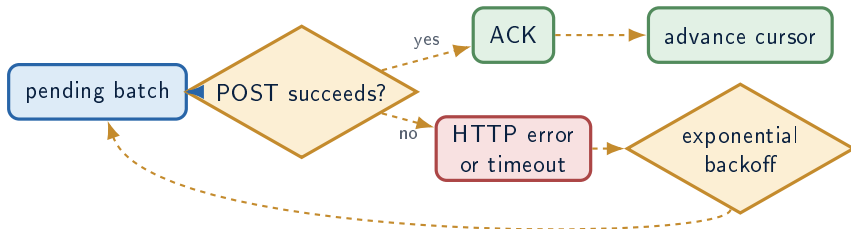
Collector Sequence: Cursor Advances After ACK



Collector State Machine

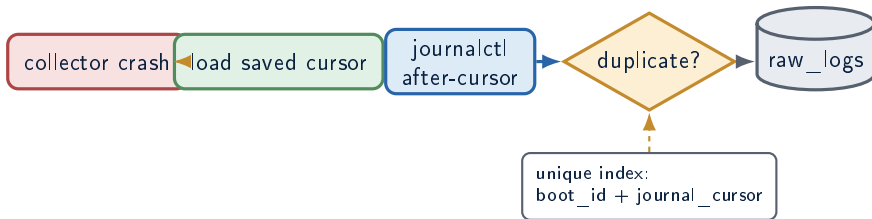


Backend Unavailable: Retry Branch



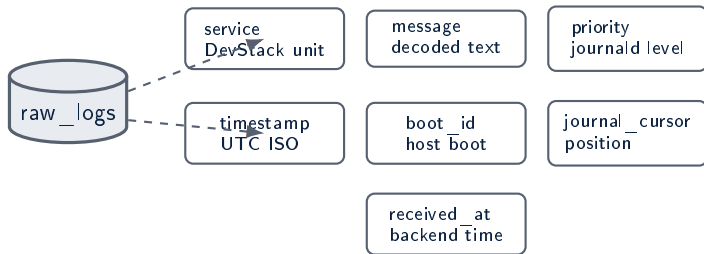
Failed batches are restored before sleeping; cursor is not advanced.

Crash Recovery and Duplicate Prevention

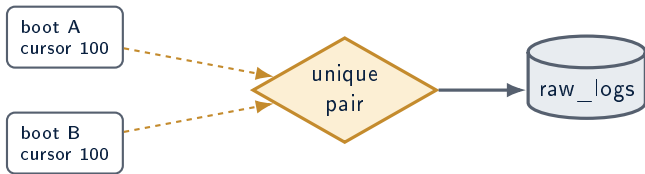


Replayed records are counted as duplicates instead of corrupting the raw collection.

raw_logs Schema as Evidence Envelope

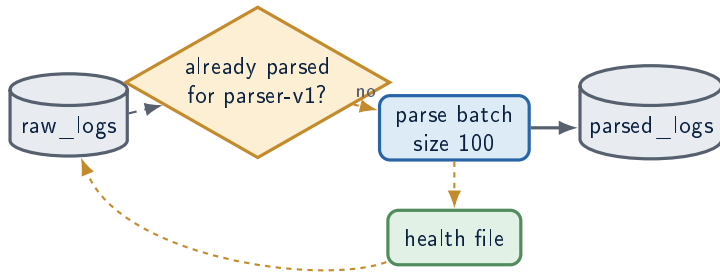


boot_id + journal_cursor Identity

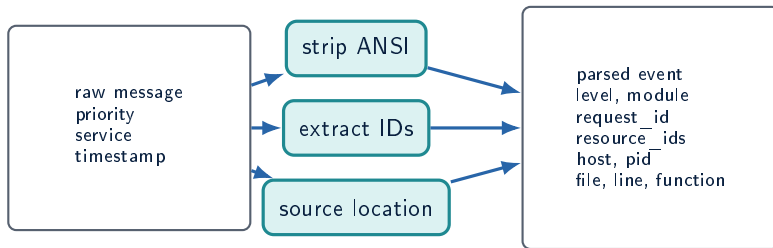


Cursor values are scoped by boot ID, so the pair is the stable journal identity.

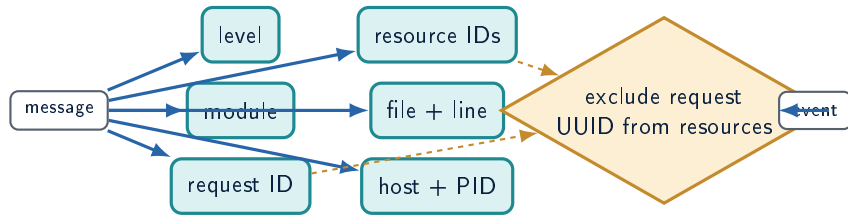
Parser Worker Lifecycle



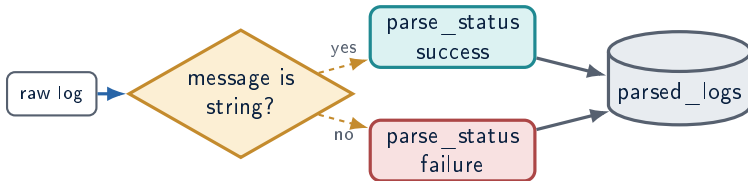
Raw Text to Structured Event



Extraction Graph

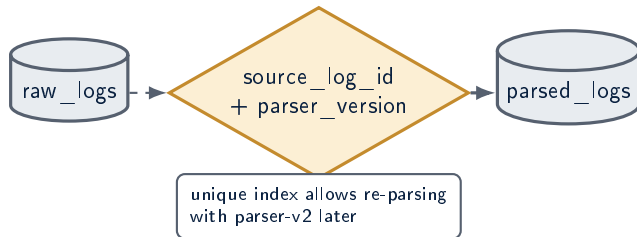


Parse Failure Preservation

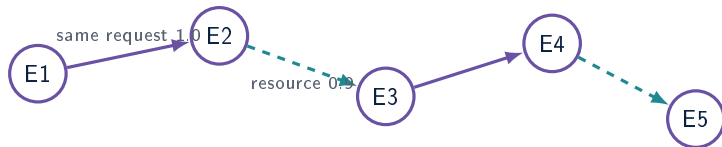


Bad records stay visible and auditable rather than disappearing.

Parser Idempotency and Versioning

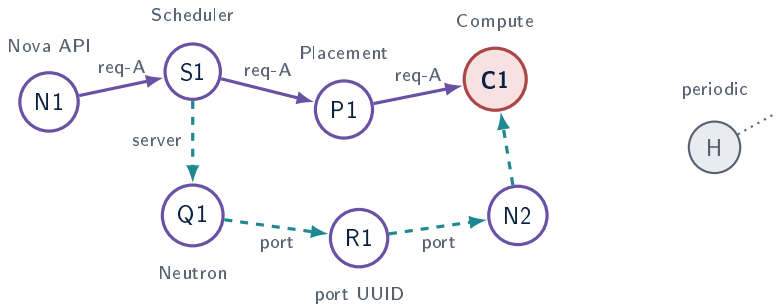


Event Nodes and Correlation Edges

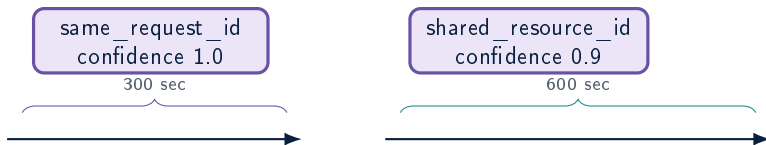


Edges are chronological: source event is earlier, target event is later.

Realistic Multi-Service Correlation Graph

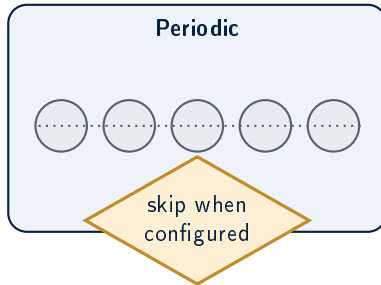
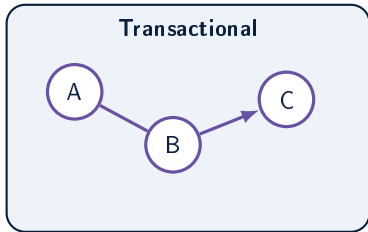


Correlation Rule Windows

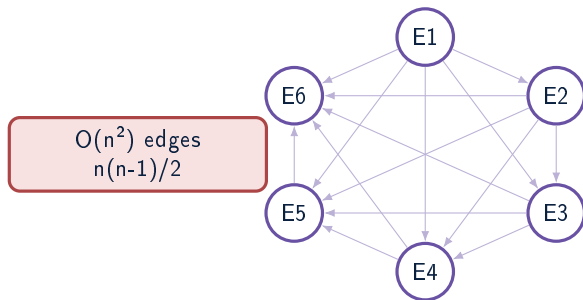


Rules fetch neighboring events inside each group window.

Transactional vs Periodic Groups



Before: Complete Pairwise Graph



Dense graphs hide signal and amplify repeated periodic IDs.

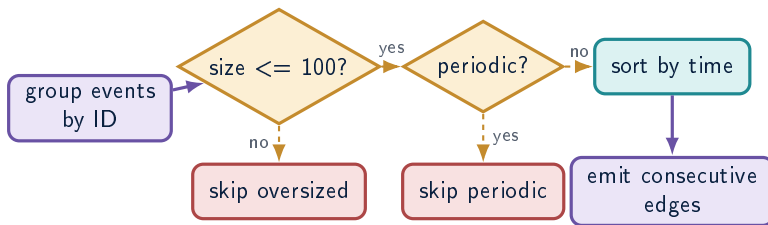
After: Consecutive-Only Chain



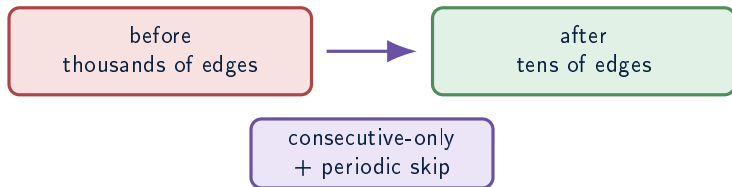
maximum $n-1$ edges

Events are sorted by timestamp and linked only to the next event in the group.

Explosion Fix Decision Flow

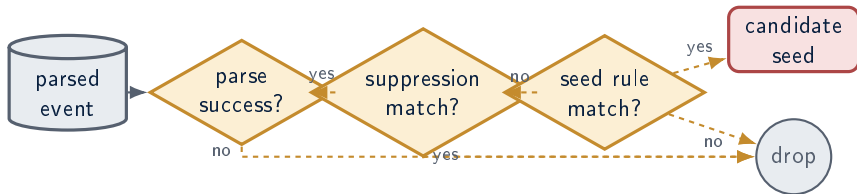


Observed Impact

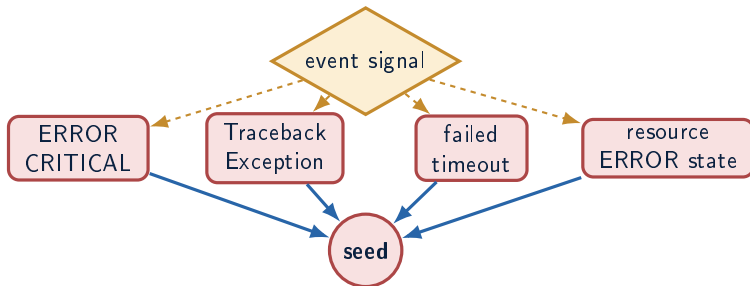


This is an engineering control against false correlation, not just a performance optimization.

Incident Seed Rule Engine

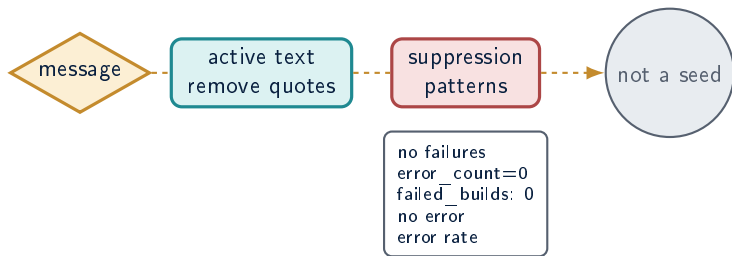


Accepted Seed Decision Tree

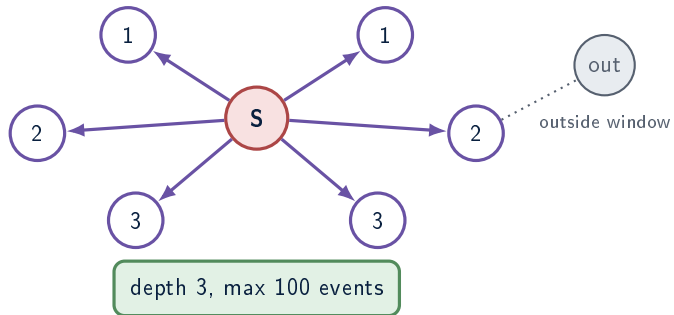


Severity is CRITICAL only when parsed level is CRITICAL; otherwise ERROR.

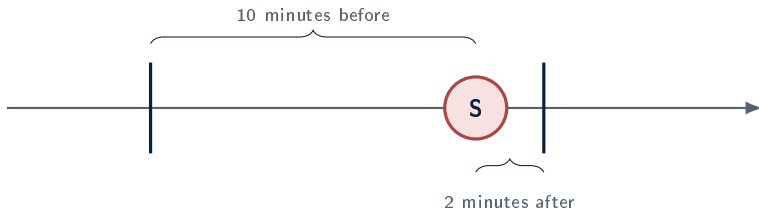
Rejected False Positives



Bounded Traversal Around Seed

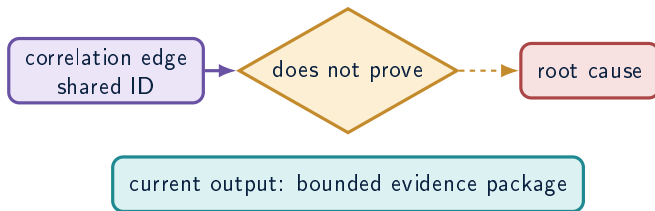


Incident Window

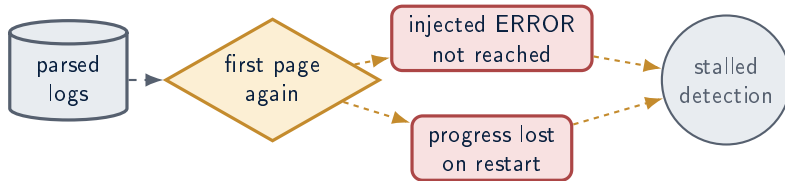


Only events inside the window can join the incident subgraph.

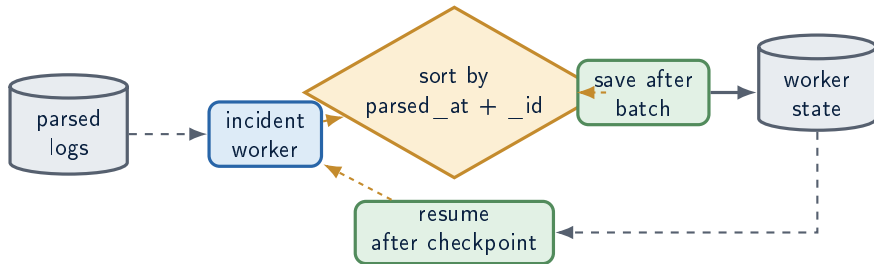
Correlation Is Not Causality



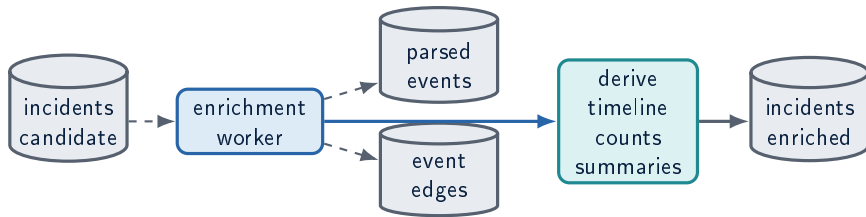
Before: Repeated First-Page Scanning



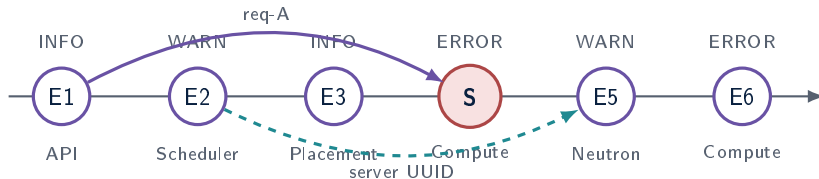
After: Persistent worker_state



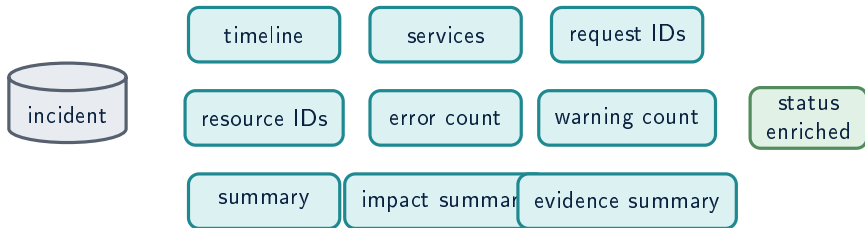
Enrichment Worker Architecture



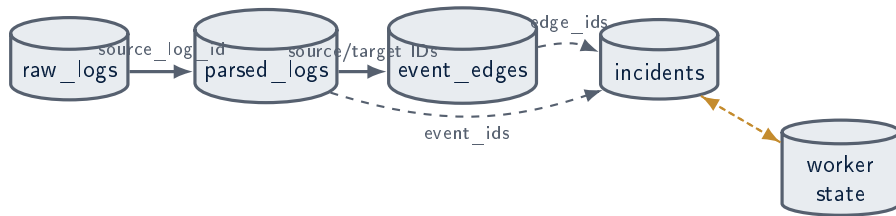
Enriched Incident Timeline



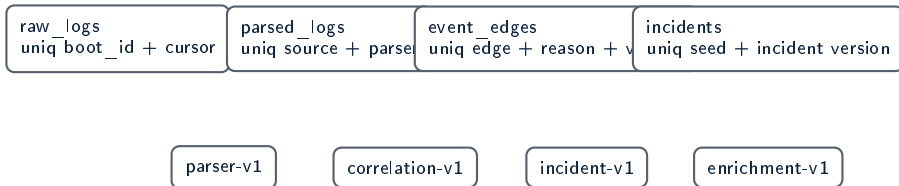
Enriched Incident Document



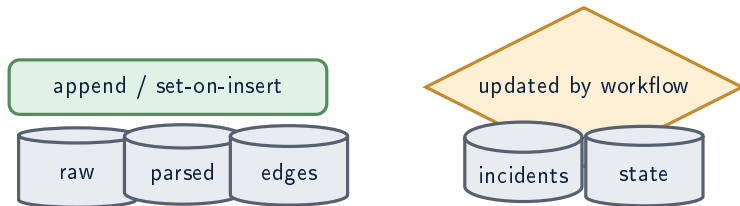
Collection Relationship Map



Indexes and Version Fields

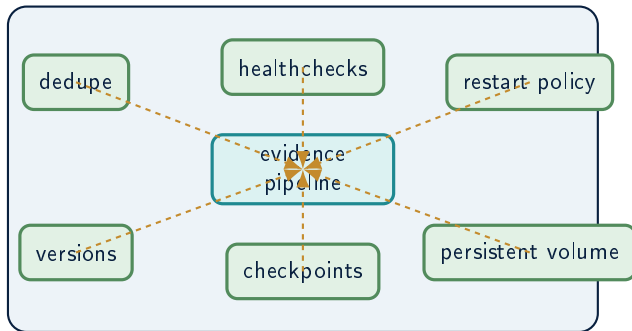


Immutable vs Mutable Collections

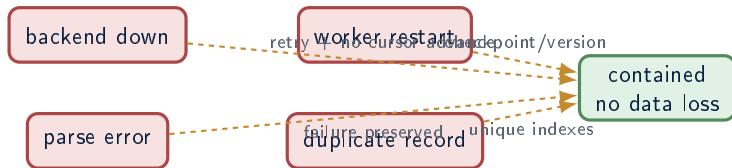


Raw evidence is preserved; incident documents mature from candidate to enriched.

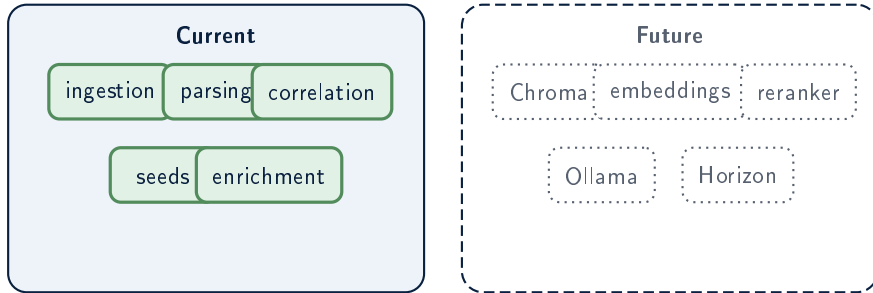
Reliability Shield



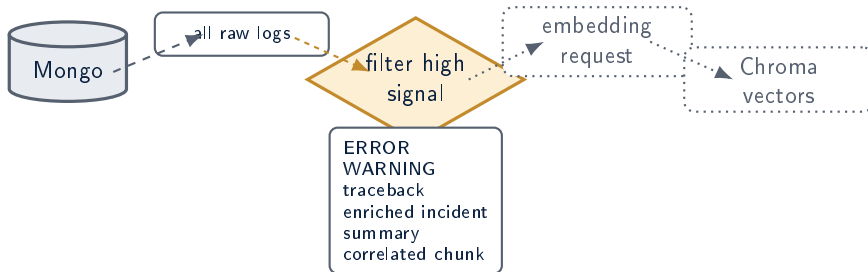
Failure Containment



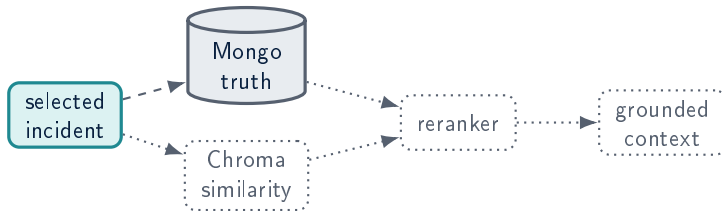
Current and Future Capability Boundary



Selective Vectorization Funnel

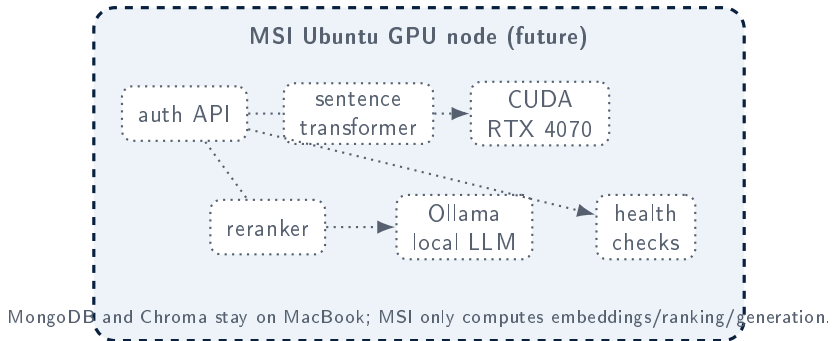


Retrieval Architecture

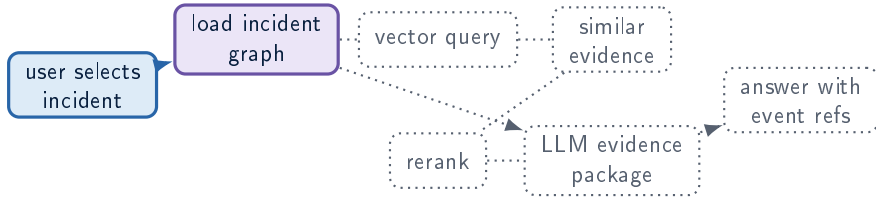


Retrieved vectors point back to MongoDB evidence IDs.

Future MSI GPU Node



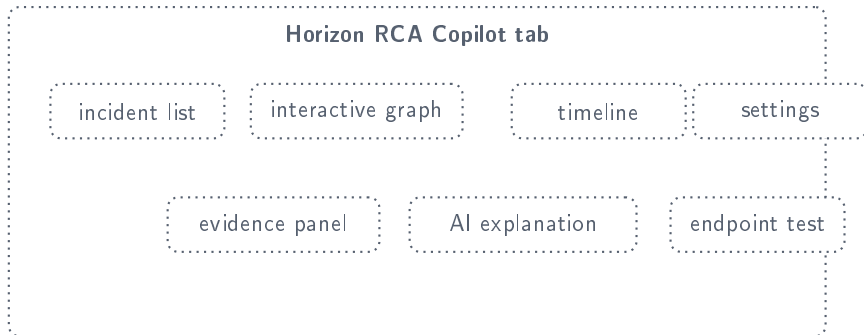
Grounded RCA Generation Flow



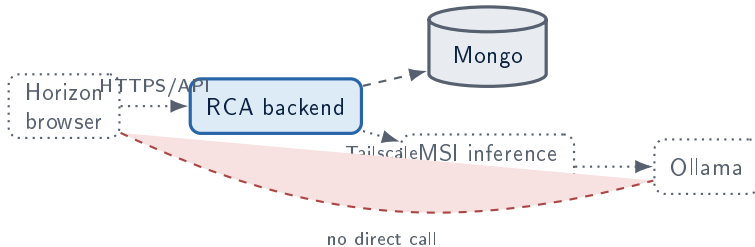
Unsupported Claim Gate



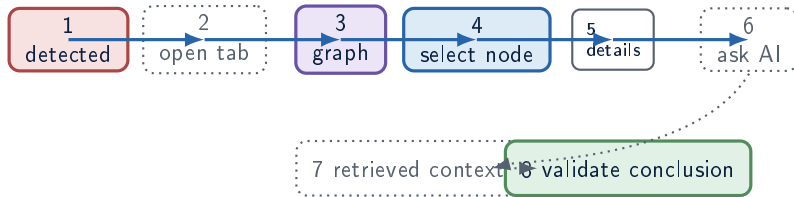
Future Horizon RCA Copilot Tab



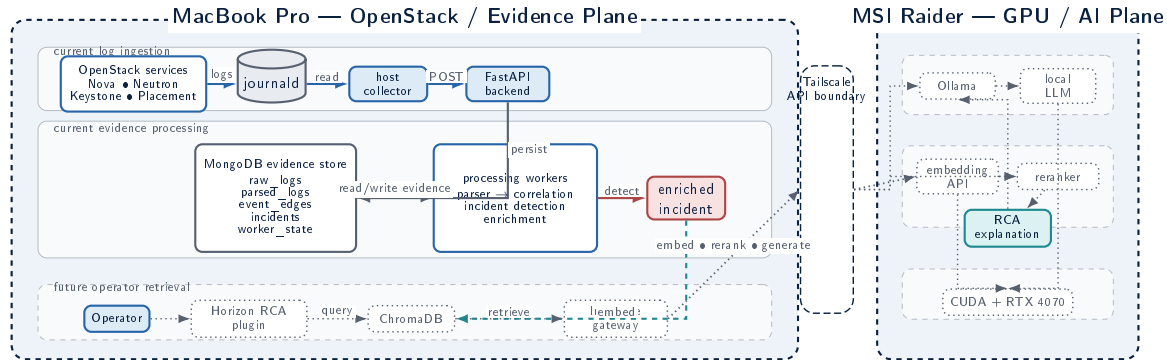
Browser-to-Backend Flow



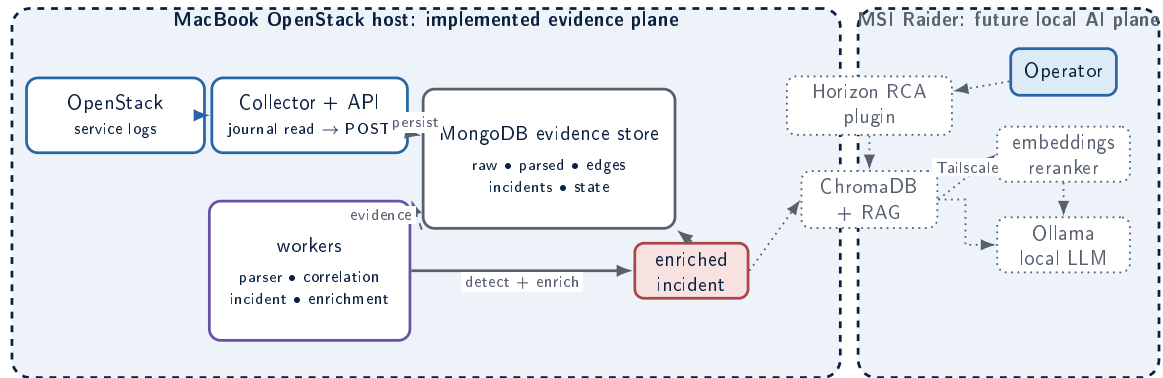
Operator Storyboard



Overall Architecture — Two-Machine RCA Copilot



Full Final Architecture



Roadmap

Completed



Future



Evidence foundation first; local AI and Horizon integration after reliable artifacts exist.

**OpenStack RCA Copilot turns
scattered service logs into a trace-
able incident evidence graph.**

graph first

evidence first

AI explains

The AI layer is valuable because the evidence layer is explicit, bounded, and auditable.